

# Supplementary Material for “**exdqlm**: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models”

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## Abstract

This supplement derives the posterior targets, full conditional distributions, variational updates, Laplace-Delta approximations, and numerical blocks used by the algorithms in **exdqlm**. It covers dynamic AL/DQLM and exAL/exDQLM state-space models, transfer-function state augmentation, static AL/exAL regression, ridge and regularized horseshoe normal-scale (RHS-NS) shrinkage priors, forecasting, diagnostics, and posterior-predictive synthesis. The notation follows the main article, and each algorithmic block is linked to the corresponding exported package function.

*Keywords:* Bayesian quantile regression, dynamic linear models, asymmetric Laplace likelihood, extended asymmetric Laplace likelihood, MCMC, variational Bayes.

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## S1. Scope and relation to the main article

The main article describes the user-facing workflow for **exdqlm**: dynamic exDQLM and DQLM fitting, transfer-function models, static exAL regression, forecasting, diagnostics, and posterior-predictive synthesis. This supplement gives the corresponding posterior targets, full conditional distributions, variational updates, Laplace-Delta approximations, ELBO components, and numerical routines used by the package. The derivations build on the AL mixture representation of [Kozumi and Kobayashi \(2011\)](#), dynamic quantile linear models ([Gonçalves, Migon, and Bastos 2017](#)), the exAL/exDQLM formulation ([Yan, Zheng, and Kottas 2025](#); [Barata, Prado, and Sansó 2022](#)), FFBS for Gaussian state-space models ([Carter and Kohn 1994](#); [Frühwirth-Schnatter 1994](#)), nonconjugate variational approximations ([Wang and Blei 2013](#)), and the RHS-NS prior ([Nishimura and Suchard 2023](#)).

Table [S1](#) gives the main navigation map. The model sections derive the fitting targets and updates, Section [S9](#) collects reusable numerical identities, and Table [S3](#) links the exported functions to the corresponding computational blocks.

The notation follows the main article. In particular,  $\mathbf{F}_t$  denotes the observation vector,  $\mathbf{G}_t$  the state evolution matrix,  $\mathbf{W}_t$  the evolution covariance,  $\boldsymbol{\theta}_t$  the dynamic state,  $p_0$  the target quantile level, and  $\gamma$  the exAL skewness parameter. Dimensions are stated when each object

is introduced.

| Main article topic                | Main article location  | Supplement location |
|-----------------------------------|------------------------|---------------------|
| exAL mixture and exDQLM hierarchy | Equations (1)–(6)      | S2–S4               |
| AL/DQLM special case              | Section 2.5            | S3                  |
| Discount-factor evolution         | Section 2.3            | S3.1, S4.1          |
| Transfer-function exDQLM          | Equations (7)–(10)     | S5                  |
| Static AL/exAL regression         | Section 2.5, Example 4 | S6–S7               |
| Forecasting                       | Equation (15)          | S8.1                |
| Diagnostics                       | Equations (11)–(14)    | S8.2                |
| Posterior-predictive synthesis    | Example 1              | S8.3                |
| Numerical blocks                  | Section 2.2            | S9                  |

Table S1: Crosswalk between the main article, this supplement, and the corresponding derivation sections.

## S2. Notation and distributional conventions

Let  $p_0 \in (0, 1)$  be the target quantile. The response is scalar. In dynamic models, observations are indexed by  $t = 1, \dots, T$ ; in static regression, observations are indexed by  $i = 1, \dots, n$ . For any positive definite matrix  $A$ ,  $|A|$  denotes its determinant.

### S2.1. Core notation

The symbols  $p_\gamma$ ,  $A_\gamma$ ,  $B_\gamma$ , and  $C_\gamma$  are deterministic functions of  $(p_0, \gamma)$ , whereas  $\pi_\gamma(\gamma)$  denotes the prior density for  $\gamma$  on  $(L, U)$ . Generic densities are denoted by  $p(\cdot)$  only when no confusion with the exAL quantile map can arise. In variational sections, expectations without an explicit subscript are taken under the current mean-field distribution  $q$ . The symbol  $\xi$  is reserved for the RHS-NS half-Cauchy auxiliary variable; the transformed coordinate for  $\gamma$  in Laplace-Delta calculations is denoted by  $z_\gamma$ .

The distributional parameterizations are also fixed throughout:  $\mathcal{N}(m, V)$  uses variance  $V$ ;

$$x \sim \text{IG}(a, b) \iff p(x) = \frac{b^a}{\Gamma(a)} x^{-a-1} \exp(-b/x), \quad x > 0, \quad (\text{S1})$$

equivalently  $x^{-1} \sim \text{Gamma}(a, b)$  under a shape-rate gamma parameterization. The exponential distribution  $\text{Exp}(\text{rate} = 1/\sigma)$  has density  $\sigma^{-1} \exp(-x/\sigma)$ , matching the main article’s mean- $\sigma$  convention for the AL and exAL mixture variables. Finally,  $\mathcal{N}^+(m, V)$  denotes  $\mathcal{N}(m, V)$  truncated to  $(0, \infty)$ . The generalized inverse Gaussian parameterization is given in (S10).

The asymmetric Laplace (AL) mixture representation used by the package is

$$v \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \quad v > 0, \quad (\text{S2})$$

$$y \mid \eta, \sigma, v \sim \mathcal{N}\{\eta + A(p_0)v, \sigma B(p_0)v\}, \quad (\text{S3})$$

| Symbol                                                            | Meaning                                                                        |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------|
| $p_0$                                                             | Target quantile level.                                                         |
| $p_\gamma = p(p_0, \gamma)$                                       | Internal exAL quantile map defined in the main article.                        |
| $A_{p_0}, B_{p_0}$                                                | AL constants $A(p_0)$ and $B(p_0)$ .                                           |
| $A_\gamma, B_\gamma, C_\gamma$                                    | exAL constants evaluated at $p_\gamma$ .                                       |
| $L, U$                                                            | Lower and upper admissible bounds for $\gamma$ .                               |
| $\pi_\gamma(\gamma)$                                              | Prior density for $\gamma$ on $(L, U)$ .                                       |
| $\boldsymbol{\theta}_t, \mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t$ | Dynamic state, observation vector, evolution matrix, and evolution covariance. |
| $v_t, s_t$                                                        | exAL latent variables in the dynamic model.                                    |
| $\boldsymbol{\beta}, \mathbf{X}$                                  | Static regression coefficients and design matrix.                              |
| $v_i, s_i$                                                        | exAL latent variables in the static model.                                     |
| $\mathcal{A}$                                                     | Active coefficient set receiving RHS-NS shrinkage.                             |
| $\mathcal{L}(q)$                                                  | Evidence lower bound.                                                          |

Table S2: Core notation used throughout the supplement. The shorthand  $p_\gamma$ ,  $A_\gamma$ ,  $B_\gamma$ , and  $C_\gamma$  refers to functions of  $(p_0, \gamma)$  defined in the main article.

where

$$A(p) = \frac{1 - 2p}{p(1 - p)}, \quad B(p) = \frac{2}{p(1 - p)}. \quad (\text{S4})$$

The AL model is the  $\gamma = 0$  special case of the exAL formulation below.

The exAL distribution is represented through an additional latent variable  $s \sim \mathcal{N}^+(0, 1)$ , a standard normal truncated to  $(0, \infty)$ . For  $\gamma \in (L, U)$ , define

$$g(\gamma) = 2\Phi(-|\gamma|) \exp(\gamma^2/2), \quad (\text{S5})$$

$$p_\gamma = \mathbb{I}(\gamma < 0) + \frac{p_0 - \mathbb{I}(\gamma < 0)}{g(\gamma)}, \quad (\text{S6})$$

$$A_\gamma = A(p_\gamma), \quad B_\gamma = B(p_\gamma), \quad C_\gamma = \{\mathbb{I}(\gamma > 0) - p_\gamma\}^{-1}. \quad (\text{S7})$$

The endpoints  $L < 0 < U$  are the roots  $g(L) = 1 - p_0$  and  $g(U) = p_0$ . Conditional on  $(v, s, \sigma, \gamma)$ ,

$$y \mid \eta, \sigma, \gamma, v, s \sim \mathcal{N}(\eta + C_\gamma \sigma |\gamma| s + A_\gamma v, \sigma B_\gamma v). \quad (\text{S8})$$

The package uses an inverse-gamma prior  $\sigma \sim \text{IG}(a_\sigma, b_\sigma)$  and a proper truncated Student- $t$  prior  $\pi_\gamma(\gamma)$  for  $\gamma$  on  $(L, U)$ . If  $m_\gamma$ ,  $s_\gamma$ , and  $\nu_\gamma$  denote its location, scale, and degrees of freedom, then

$$\pi_\gamma(\gamma) = \frac{s_\gamma^{-1} t_{\nu_\gamma}\{(\gamma - m_\gamma)/s_\gamma\}}{T_{\nu_\gamma}\{(U - m_\gamma)/s_\gamma\} - T_{\nu_\gamma}\{(L - m_\gamma)/s_\gamma\}} \mathbb{I}(L < \gamma < U), \quad (\text{S9})$$

where  $t_\nu(\cdot)$  and  $T_\nu(\cdot)$  are the standard Student- $t$  density and distribution function.

The generalized inverse Gaussian distribution is parameterized as

$$x \sim \text{GIG}(\lambda, \chi, \psi_{\text{GIG}}) \iff p(x) \propto x^{\lambda-1} \exp\left\{-\frac{1}{2}\left(\frac{\chi}{x} + \psi_{\text{GIG}} x\right)\right\}, \quad x > 0. \quad (\text{S10})$$

For  $r \in \mathbb{R}$ ,

$$\mathbb{E}(x^r) = \left(\frac{\chi}{\psi_{\text{GIG}}}\right)^{r/2} \frac{K_{\lambda+r}(\sqrt{\chi\psi_{\text{GIG}}})}{K_\lambda(\sqrt{\chi\psi_{\text{GIG}}})}, \quad (\text{S11})$$

where  $K_\nu(\cdot)$  is the modified Bessel function of the second kind.

### S3. Dynamic DQLM under the AL likelihood

#### S3.1. Model hierarchy and full joint

Let  $\boldsymbol{\theta}_t \in \mathbb{R}^q$  be the latent state,  $\mathbf{F}_t \in \mathbb{R}^q$  the observation vector,  $\mathbf{G}_t \in \mathbb{R}^{q \times q}$  the evolution matrix, and  $\mathbf{W}_t \in \mathbb{R}^{q \times q}$  a positive semidefinite evolution covariance. The dynamic DQLM is

$$\boldsymbol{\theta}_0 \sim \mathcal{N}(\mathbf{m}_0, \mathbf{C}_0), \quad \mathbf{m}_0 \in \mathbb{R}^q, \quad \mathbf{C}_0 \in \mathbb{R}^{q \times q}, \quad (\text{S12})$$

$$\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} \sim \mathcal{N}(\mathbf{G}_t \boldsymbol{\theta}_{t-1}, \mathbf{W}_t), \quad (\text{S13})$$

$$v_t \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \quad (\text{S14})$$

$$y_t \mid \boldsymbol{\theta}_t, \sigma, v_t \sim \mathcal{N}(\mathbf{F}_t^\top \boldsymbol{\theta}_t + A_{p_0} v_t, \sigma B_{p_0} v_t), \quad (\text{S15})$$

where  $A_{p_0} = A(p_0)$  and  $B_{p_0} = B(p_0)$ . With  $\mathbf{v} = (v_1, \dots, v_T)^\top$  and  $\boldsymbol{\theta}_{0:T} = (\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_T)$ , the posterior target is

$$p(\boldsymbol{\theta}_{0:T}, \sigma, \mathbf{v} \mid \mathbf{y}) \propto p(\sigma) p(\boldsymbol{\theta}_0) \prod_{t=1}^T p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \prod_{t=1}^T p(v_t \mid \sigma) \prod_{t=1}^T p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t). \quad (\text{S16})$$

Equivalently, conditional on  $(\sigma, \mathbf{v})$ , the pseudo-observation  $\tilde{y}_t = y_t - A_{p_0} v_t$  satisfies

$$\tilde{y}_t = \mathbf{F}_t^\top \boldsymbol{\theta}_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma B_{p_0} v_t). \quad (\text{S17})$$

Thus the conditional state posterior is a Gaussian state-space posterior. The package implements the evolution variance through the discount-factor state-space construction used by `exdq1mMCMC()` and `exdq1mLDVB()`; the notation above writes the induced covariance generically as  $\mathbf{W}_t$ . The derivations condition on the sequence  $\mathbf{W}_{1:T}$  induced by the discount-factor construction in the main article. Componentwise discounting changes the block structure of  $\mathbf{W}_t$ ; once this sequence is fixed within a Gaussian pseudo-state update, the FFBS and LDVB state updates have the forms below.

#### S3.2. MCMC updates

One Gibbs sweep for the dynamic DQLM uses the following full conditionals.

**State path.** Given  $(\sigma, \mathbf{v})$ , sample the state path  $\boldsymbol{\theta}_{0:T}$  exactly by applying forward-filtering backward-sampling (FFBS) to (S17).

**Scale.** Let  $r_t = y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t$ . Then

$$\sigma \mid \boldsymbol{\theta}_{0:T}, \mathbf{v}, \mathbf{y} \sim \text{IG} \left( a_\sigma + \frac{3T}{2}, b_\sigma + \sum_{t=1}^T v_t + \sum_{t=1}^T \frac{(r_t - A_{p_0} v_t)^2}{2B_{p_0} v_t} \right). \quad (\text{S18})$$

**Latent AL variables.** For  $t = 1, \dots, T$ ,

$$v_t \mid \boldsymbol{\theta}_t, \sigma, y_t \sim \text{GIG} \left( \frac{1}{2}, \frac{r_t^2}{\sigma B_{p_0}}, \frac{A_{p_0}^2}{\sigma B_{p_0}} + \frac{2}{\sigma} \right). \quad (\text{S19})$$

**Algorithm S1 (dynamic DQLM MCMC).** The MCMC kernel leaves the augmented posterior in (S16) invariant; posterior summaries are Monte Carlo estimates from the retained finite chain.

- (1) Initialize  $\boldsymbol{\theta}_{0:T}$ ,  $\mathbf{v}$ , and  $\sigma$ .
- (2) For  $m = 1, \dots, N_{\text{burn}} + N_{\text{mcmc}}$ :
  - (a) sample  $\boldsymbol{\theta}_{0:T}$  by FFBS using (S17);
  - (b) sample  $\sigma$  from (S18);
  - (c) sample each  $v_t$  from (S19);
  - (d) retain draws after burn-in.
- (3) Return posterior samples and filtered/smoothed state summaries.

### S3.3. Mean-field VB and ELBO

The dynamic DQLM VB approximation uses

$$q(\boldsymbol{\theta}_{0:T}, \sigma, \mathbf{v}) = q(\boldsymbol{\theta}_{0:T})q(\sigma) \prod_{t=1}^T q(v_t), \quad (\text{S20})$$

with  $q(\sigma) = \text{IG}(a_q, b_q)$  and  $q(v_t) = \text{GIG}(1/2, \chi_t, \psi_t)$ . Define

$$\kappa = \mathbb{E}_q(\sigma^{-1}) = \frac{a_q}{b_q}, \quad \ell_t = \mathbb{E}_q(v_t^{-1}), \quad \bar{v}_t = \mathbb{E}_q(v_t). \quad (\text{S21})$$

The state factor  $q(\boldsymbol{\theta}_{0:T})$  is a joint Gaussian smoothing distribution, not a product over time. The latent-variable updates require only its marginal means and variances. The state factor is the Gaussian posterior of the auxiliary model

$$y_t^* = \mathbf{F}_t^\top \boldsymbol{\theta}_t + \epsilon_t^*, \quad \epsilon_t^* \sim \mathcal{N}(0, R_t^*), \quad y_t^* = y_t - \frac{A_{p_0}}{\ell_t}, \quad R_t^* = \frac{B_{p_0}}{\kappa \ell_t}. \quad (\text{S22})$$

After Kalman smoothing, write  $q(\boldsymbol{\theta}_t) = \mathcal{N}(\mathbf{m}_{t|T}, \mathbf{C}_{t|T})$  and define

$$\bar{r}_t = y_t - \mathbf{F}_t^\top \mathbf{m}_{t|T}, \quad (\text{S23})$$

$$\bar{r}_t^2 = \bar{r}_t^2 + \mathbf{F}_t^\top \mathbf{C}_{t|T} \mathbf{F}_t. \quad (\text{S24})$$

The remaining coordinate-ascent variational inference (CAVI) updates are

$$q(v_t) = \text{GIG}\left(\frac{1}{2}, \frac{\kappa}{B_{p_0}} \bar{r}_t^2, \kappa \left(2 + \frac{A_{p_0}^2}{B_{p_0}}\right)\right), \quad (\text{S25})$$

$$q(\sigma) = \text{IG}(a_q, b_q), \quad a_q = a_\sigma + \frac{3T}{2}, \quad (\text{S26})$$

$$b_q = b_\sigma + \sum_{t=1}^T \bar{v}_t + \frac{1}{2B_{p_0}} \sum_{t=1}^T \left\{ \ell_t \bar{r}_t^2 - 2A_{p_0} \bar{r}_t + A_{p_0}^2 \bar{v}_t \right\}. \quad (\text{S27})$$

**Algorithm S2 (dynamic DQLM LDVB).**

- (1) Initialize the variational parameters for  $q(\boldsymbol{\theta}_{0:T})$ ,  $q(\sigma)$ , and  $q(v_t)$ .
- (2) Repeat until convergence:
  - (a) update  $q(\boldsymbol{\theta}_{0:T})$  by Kalman smoothing under (S22);
  - (b) update  $q(v_t)$  for  $t = 1, \dots, T$ ;
  - (c) update  $q(\sigma)$  using (S27);
  - (d) monitor moment changes and, when enabled,  $\mathcal{L}(q)$ .
- (3) Draw approximate posterior samples from the fitted variational factors when requested by `n.samp`.

The ELBO is

$$\mathcal{L}(q) = \mathbb{E}_q\{\log p(\mathbf{y}, \boldsymbol{\theta}_{0:T}, \sigma, \mathbf{v})\} - \mathbb{E}_q\{\log q(\boldsymbol{\theta}_{0:T}, \sigma, \mathbf{v})\} \quad (\text{S28})$$

$$= \mathcal{L}_\theta + \mathbb{E}_q\{\log p(\sigma)\} + \sum_{t=1}^T \mathbb{E}_q\{\log p(v_t \mid \sigma)\} + \sum_{t=1}^T \mathbb{E}_q\{\log p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t)\} \quad (\text{S29})$$

$$+ H\{q(\sigma)\} + \sum_{t=1}^T H\{q(v_t)\}. \quad (\text{S30})$$

The state-path term  $\mathcal{L}_\theta$  is evaluated from the auxiliary Gaussian state-space model in (S22) through the Kalman marginal likelihood identity

$$\mathcal{L}_\theta = \log p(\mathbf{y}^*) - \mathbb{E}_q\{\log p(\mathbf{y}^* \mid \boldsymbol{\theta}_{0:T})\}, \quad (\text{S31})$$

where  $p(\mathbf{y}^*)$  is the marginal likelihood of the auxiliary Gaussian state-space model. This identity is equivalent to  $\mathbb{E}_q\{\log p(\boldsymbol{\theta}_{0:T})\} + H\{q(\boldsymbol{\theta}_{0:T})\}$  for the current state update.

The remaining terms in (S30) are computable from the current variational moments. Let  $\bar{\ell}_\sigma = \mathbb{E}(\log \sigma)$  and  $\bar{\ell}_{v,t} = \mathbb{E}(\log v_t)$ . Then

$$\mathbb{E}_q\{\log p(\sigma)\} = a_\sigma \log b_\sigma - \log \Gamma(a_\sigma) - (a_\sigma + 1)\bar{\ell}_\sigma - b_\sigma \kappa, \quad (\text{S32})$$

$$\sum_{t=1}^T \mathbb{E}_q\{\log p(v_t \mid \sigma)\} = -T\bar{\ell}_\sigma - \kappa \sum_{t=1}^T \nu_t, \quad (\text{S33})$$

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}_q\{\log p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t)\} &= -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log B_{p_0} - \frac{T}{2} \bar{\ell}_\sigma - \frac{1}{2} \sum_{t=1}^T \bar{\ell}_{v,t} \\ &\quad - \frac{\kappa}{2B_{p_0}} \sum_{t=1}^T \left\{ \ell_t \bar{r}_t^2 - 2A_{p_0} \bar{r}_t + A_{p_0}^2 \bar{v}_t \right\}. \end{aligned} \quad (\text{S34})$$

The entropy terms are given by (S146) for  $q(\sigma)$  and by (S147) for the  $q(v_t)$  factors.

## S4. Dynamic exDQLM under the exAL likelihood

#### S4.1. Model hierarchy and full joint

The dynamic exDQLM replaces the AL observation in (S15) with the exAL augmentation

$$v_t \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \quad s_t \sim \mathcal{N}^+(0, 1), \quad (\text{S35})$$

$$y_t \mid \boldsymbol{\theta}_t, \sigma, \gamma, v_t, s_t \sim \mathcal{N}\left(\mathbf{F}_t^\top \boldsymbol{\theta}_t + C_\gamma \sigma |\gamma| s_t + A_\gamma v_t, \sigma B_\gamma v_t\right), \quad (\text{S36})$$

with the same state evolution as in Section S3. The complete posterior target is

$$\begin{aligned} p(\boldsymbol{\theta}_{0:T}, \mathbf{v}, \mathbf{s}, \sigma, \gamma \mid \mathbf{y}) &\propto p(\boldsymbol{\theta}_0) \prod_{t=1}^T p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \prod_{t=1}^T p(y_t \mid \boldsymbol{\theta}_t, v_t, s_t, \sigma, \gamma) \\ &\times \prod_{t=1}^T p(v_t \mid \sigma) \prod_{t=1}^T p(s_t) p(\sigma) \pi_\gamma(\gamma). \end{aligned} \quad (\text{S37})$$

Conditional on  $(\mathbf{v}, \mathbf{s}, \sigma, \gamma)$ ,

$$\tilde{y}_t = y_t - A_\gamma v_t - C_\gamma \sigma |\gamma| s_t, \quad R_t^y = \sigma B_\gamma v_t, \quad (\text{S38})$$

which again gives a linear Gaussian state-space model for  $\boldsymbol{\theta}_{0:T}$ .

#### S4.2. MCMC updates

The default package path samples  $\sigma$  from its exact conditional given  $\gamma$  and then updates  $\gamma$  by bounded univariate slice sampling. Joint random-walk alternatives are also implemented, but the following target is the default exact-kernel path.

**Latent  $v_t$ .** Let  $r_t = y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t - C_\gamma \sigma |\gamma| s_t$ . Then

$$v_t \mid \cdot \sim \text{GIG}\left(\frac{1}{2}, \frac{r_t^2}{\sigma B_\gamma}, \frac{A_\gamma^2}{\sigma B_\gamma} + \frac{2}{\sigma}\right). \quad (\text{S39})$$

**Latent  $s_t$ .** Let  $y_t^\circ = y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t$  and  $D_\gamma = C_\gamma \sigma |\gamma|$ . Then

$$s_t \mid \cdot \sim \mathcal{N}^+(m_{s,t}, V_{s,t}), \quad V_{s,t} = \left(1 + \frac{D_\gamma^2}{\sigma B_\gamma v_t}\right)^{-1}, \quad m_{s,t} = V_{s,t} \frac{D_\gamma y_t^\circ}{\sigma B_\gamma v_t}. \quad (\text{S40})$$

**State path.** Given  $(\mathbf{v}, \mathbf{s}, \sigma, \gamma)$ , sample  $\boldsymbol{\theta}_{0:T}$  by FFBS using (S38).

**Scale.** For fixed  $\gamma$ , let  $y_t^\circ = y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t$  and  $c_{\gamma,t} = C_\gamma |\gamma| s_t$ . Then

$$\sigma \mid \cdot \sim \text{GIG}(\lambda_\sigma, \chi_\sigma, \psi_\sigma), \quad (\text{S41})$$

where

$$\lambda_\sigma = -\left(a_\sigma + \frac{3T}{2}\right), \quad (\text{S42})$$

$$\chi_\sigma = 2b_\sigma + 2 \sum_{t=1}^T v_t + \sum_{t=1}^T \frac{(y_t^\circ)^2}{B_\gamma v_t}, \quad (\text{S43})$$

$$\psi_\sigma = \sum_{t=1}^T \frac{c_{\gamma,t}^2}{B_\gamma v_t}. \quad (\text{S44})$$

**Skewness.** The conditional kernel for  $\gamma$  is

$$\pi(\gamma \mid \cdot) \propto \pi_\gamma(\gamma) \prod_{t=1}^T (\sigma B_\gamma v_t)^{-1/2} \exp \left[ - \sum_{t=1}^T \frac{\{y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t - C_\gamma \sigma |\gamma| s_t\}^2}{2\sigma B_\gamma v_t} \right], \quad (\text{S45})$$

for  $\gamma \in (L, U)$ . The default sampler applies the bounded slice update in Section S9.2 directly to this kernel.

**Algorithm S3 (dynamic exDQLM MCMC).** The default  $(\sigma, \gamma)$  update consists of an exact GIG update for  $\sigma$  conditional on  $\gamma$ , followed by a bounded univariate slice update for  $\gamma$ . Joint random-walk alternatives are implemented, but they are not the default kernel summarized here.

- (1) Initialize  $\boldsymbol{\theta}_{0:T}$ ,  $\mathbf{v}$ ,  $\mathbf{s}$ ,  $\sigma$ , and  $\gamma$ .
- (2) For each MCMC iteration:
  - (a) sample  $v_t$  from (S39),  $t = 1, \dots, T$ ;
  - (b) sample  $s_t$  from (S40),  $t = 1, \dots, T$ ;
  - (c) sample  $\boldsymbol{\theta}_{0:T}$  by FFBS using (S38);
  - (d) sample  $\sigma$  from (S41);
  - (e) sample  $\gamma$  on  $(L, U)$  using (S45);
  - (f) retain draws after burn-in.
- (3) Return posterior samples and state summaries.

Setting  $\gamma = 0$  drops the  $s_t$  block and reduces the sampler to the AL/DQLM kernel in Algorithm S1.

### S4.3. Laplace-Delta VB approximation

The mean-field family is

$$q(\boldsymbol{\theta}_{0:T}, \mathbf{v}, \mathbf{s}, \sigma, \gamma) = q(\boldsymbol{\theta}_{0:T}) \prod_{t=1}^T q(v_t) \prod_{t=1}^T q(s_t) q(\sigma, \gamma). \quad (\text{S46})$$

The conjugate factors are updated conditionally on moments from  $q(\sigma, \gamma)$ . Define

$$\kappa_1 = \mathbb{E}\{1/(\sigma B_\gamma)\}, \quad \kappa_2 = \mathbb{E}\{A_\gamma/(\sigma B_\gamma)\}, \quad \kappa_3 = \mathbb{E}\{A_\gamma^2/(\sigma B_\gamma)\}, \quad (\text{S47})$$

$$\kappa_4 = \mathbb{E}\{C_\gamma |\gamma| / B_\gamma\}, \quad \kappa_5 = \mathbb{E}\{\sigma C_\gamma^2 \gamma^2 / B_\gamma\}, \quad \kappa_6 = \mathbb{E}\{A_\gamma C_\gamma |\gamma| / B_\gamma\}. \quad (\text{S48})$$

Let  $m_{1/v,t} = \mathbb{E}(v_t^{-1})$ ,  $m_{s,t} = \mathbb{E}(s_t)$ , and  $m_{s^2,t} = \mathbb{E}(s_t^2)$ . The state factor is Gaussian and is computed by Kalman smoothing with observation precision

$$\phi_t = \kappa_1 m_{1/v,t} \quad (\text{S49})$$

and pseudo-response

$$y_t^{\text{VB}} = \frac{y_t \phi_t - \kappa_2 - \kappa_4 m_{s,t} m_{1/v,t}}{\phi_t}. \quad (\text{S50})$$



If  $\eta_t = \mathbf{F}_t^\top \boldsymbol{\theta}_t$ , write  $m_{\eta,t} = \mathbb{E}(\eta_t)$ ,  $S_{\eta,t} = \mathbb{V}(\eta_t)$ ,  $m_{\Delta,t} = y_t - m_{\eta,t}$ , and  $S_{\Delta,t} = m_{\Delta,t}^2 + S_{\eta,t}$ . Then

$$q(v_t) = \text{GIG}\left(\frac{1}{2}, \kappa_1 S_{\Delta,t} - 2\kappa_4 m_{\Delta,t} m_{s,t} + \kappa_5 m_{s^2,t}, \kappa_3 + 2\mathbb{E}(\sigma^{-1})\right), \quad (\text{S51})$$

$$q(s_t) = \mathcal{N}^+(m_{s,t}^{\text{VB}}, V_{s,t}^{\text{VB}}), \quad (\text{S52})$$

$$V_{s,t}^{\text{VB}} = \left(1 + \kappa_5 m_{1/v,t}\right)^{-1}, \quad (\text{S53})$$

$$m_{s,t}^{\text{VB}} = V_{s,t}^{\text{VB}} \left(\kappa_4 m_{\Delta,t} m_{1/v,t} - \kappa_6\right). \quad (\text{S54})$$

The nonconjugate factor  $q(\sigma, \gamma)$  is approximated by a Laplace-Delta step on transformed coordinates

$$\ell_\sigma = \log \sigma, \quad z_\gamma = \text{logit}\left(\frac{\gamma - L}{U - L}\right). \quad (\text{S55})$$

The transformed objective is the expected complete log-joint plus the Jacobian,

$$h_z(\ell_\sigma, z_\gamma) = \mathbb{E}_{-\sigma, \gamma} \{\log p(\mathbf{y}, \boldsymbol{\theta}_{0:T}, \mathbf{v}, \mathbf{s}, \sigma, \gamma)\} + \ell_\sigma + \log(U - L) + \log r + \log(1 - r), \quad r = \text{expit}(z_\gamma). \quad (\text{S56})$$

The package stores the transformed mode, the approximate covariance matrix, Delta-method moments, and ELBO components.

#### Algorithm S4 (dynamic exDQLM LDVB).

- (1) Initialize variational parameters for  $q(\boldsymbol{\theta}_{0:T})$ ,  $q(v_t)$ ,  $q(s_t)$ , and  $q(\sigma, \gamma)$ .
- (2) Repeat until convergence:
  - (a) update  $q(\boldsymbol{\theta}_{0:T})$  by Kalman smoothing under (S50);
  - (b) update  $q(v_t)$  and  $q(s_t)$  using (S54);
  - (c) update  $q(\sigma, \gamma)$  by the Laplace-Delta approximation to (S56);
  - (d) monitor moment changes and, when enabled,  $\mathcal{L}(q)$ .
- (3) Draw approximate posterior samples from the fitted variational factors when requested by `n.samp`.

In the AL special case, the nonconjugate  $(\sigma, \gamma)$  block is replaced by the inverse-gamma  $q(\sigma)$  update in Algorithm S2.

**ELBO.** The LDVB ELBO has the block decomposition

$$\mathcal{L}(q) = \mathcal{L}_{\text{like}} + \mathcal{L}_\theta + \mathcal{L}_v + \mathcal{L}_s + \mathcal{L}_{\sigma\gamma} + \sum_{t=1}^T H\{q(v_t)\} + \sum_{t=1}^T H\{q(s_t)\} + H\{q(\sigma, \gamma)\}. \quad (\text{S57})$$

The state block is

$$\mathcal{L}_\theta = \mathbb{E}_q \left\{ \log p(\boldsymbol{\theta}_0) + \sum_{t=1}^T \log p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \right\} + H\{q(\boldsymbol{\theta}_{0:T})\}. \quad (\text{S58})$$

As in the AL case, this Gaussian state-space contribution can be evaluated by the auxiliary Kalman normalizing-constant identity for the current pseudo-response and variance.

Let  $\bar{\ell}_\sigma = \mathbb{E}(\log \sigma)$ ,  $\bar{\ell}_B = \mathbb{E}(\log B_\gamma)$ ,  $\bar{\ell}_\pi = \mathbb{E}\{\log \pi_\gamma(\gamma)\}$ ,  $m_{v,t} = \mathbb{E}(v_t)$ ,  $m_{1/v,t} = \mathbb{E}(v_t^{-1})$ ,  $m_{\log v,t} = \mathbb{E}(\log v_t)$ ,  $m_{s,t} = \mathbb{E}(s_t)$ , and  $m_{s^2,t} = \mathbb{E}(s_t^2)$ . Define

$$Q_t = \kappa_1 S_{\Delta,t} m_{1/v,t} - 2\kappa_2 m_{\Delta,t} + \kappa_3 m_{v,t} - 2\kappa_4 m_{\Delta,t} m_{s,t} m_{1/v,t} + \kappa_5 m_{s^2,t} m_{1/v,t} + 2\kappa_6 m_{s,t}. \quad (\text{S59})$$

Then the likelihood, latent-prior, and nonconjugate-prior blocks are

$$\mathcal{L}_{\text{like}} = -\frac{T}{2} \log(2\pi) - \frac{T}{2} (\bar{\ell}_\sigma + \bar{\ell}_B) - \frac{1}{2} \sum_{t=1}^T m_{\log v,t} - \frac{1}{2} \sum_{t=1}^T Q_t, \quad (\text{S60})$$

$$\mathcal{L}_v = -T\bar{\ell}_\sigma - \mathbb{E}(\sigma^{-1}) \sum_{t=1}^T m_{v,t}, \quad (\text{S61})$$

$$\mathcal{L}_s = T \log 2 - \frac{T}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T m_{s^2,t}, \quad (\text{S62})$$

$$\mathcal{L}_{\sigma\gamma} = a_\sigma \log b_\sigma - \log \Gamma(a_\sigma) - (a_\sigma + 1) \bar{\ell}_\sigma - b_\sigma \mathbb{E}(\sigma^{-1}) + \bar{\ell}_\pi. \quad (\text{S63})$$

The GIG and truncated-normal entropy terms are given by (S147) and (S148). The  $(\sigma, \gamma)$  entropy and the smooth expectations in the  $\kappa_j$ ,  $\bar{\ell}_B$ ,  $\bar{\ell}_\pi$ , and  $\mathbb{E}(\sigma^{-1})$  terms are evaluated under the Laplace-Delta approximation in (S150) and (S153).

## S5. Transfer-function exDQLM by state augmentation

Transfer-function inference uses the same AL/exAL likelihood and latent-variable updates as the dynamic models above. The only change is the state-space construction. Let  $x_t \in \mathbb{R}^k$  be the input vector,  $\zeta_t$  the transfer state, and  $\psi_t \in \mathbb{R}^k$  the input-effect state. For fixed transfer rate  $\lambda$ , the augmented model uses

$$\tilde{\mathbf{F}}_t^\top = (\mathbf{F}_t^\top, 1, 0_k^\top), \quad \tilde{\boldsymbol{\theta}}_t^\top = (\boldsymbol{\theta}_t^\top, \zeta_t, \psi_t^\top), \quad (\text{S64})$$

$$\tilde{\mathbf{G}}_t = \text{blockdiag} \left\{ \mathbf{G}_t, \begin{pmatrix} \lambda & x_t^\top \\ 0_k & I_k \end{pmatrix} \right\}, \quad \tilde{\mathbf{W}}_t = \text{blockdiag} \{ \mathbf{W}_t^\theta, \omega_t, W_t^\psi \}. \quad (\text{S65})$$

The prior for  $(\zeta_0, \psi_0^\top)^\top$  is  $\mathcal{N}(m_0^{\text{tf}}, C_0^{\text{tf}})$  and is appended to the base state prior. Conditional on  $\lambda$ , the transfer-function model is an ordinary exDQLM on the augmented state  $\tilde{\boldsymbol{\theta}}_t$ . No separate transfer-function posterior derivation is needed. Algorithms S3 and S4 apply after replacing  $(\mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t, \boldsymbol{\theta}_t)$  by  $(\tilde{\mathbf{F}}_t, \tilde{\mathbf{G}}_t, \tilde{\mathbf{W}}_t, \tilde{\boldsymbol{\theta}}_t)$ . The package inputs `X`, `lam`, `tf.df`, `tf.m0`, and `tf.c0` correspond to  $x_t$ ,  $\lambda$ , the transfer discount factors,  $m_0^{\text{tf}}$ , and  $C_0^{\text{tf}}$ ; `median.kt` is returned as a summary of the fitted transfer response.

### Algorithm S5 (transfer-function exDQLM).

- (1) Construct  $\tilde{\mathbf{F}}_t$ ,  $\tilde{\mathbf{G}}_t$ ,  $\tilde{\mathbf{W}}_t$ , and  $\tilde{\boldsymbol{\theta}}_t$  from (S64)–(S65).
- (2) Set the augmented initial prior.

- (3) Run the selected fitting routine: MCMC uses Algorithm S3 and LDVB uses Algorithm S4.
- (4) Return summaries for both the base state and the transfer states.

The transfer-function extension changes only the state-space matrices; the observation likelihood and latent-variable updates are unchanged.

## S6. Static AL and exAL regression with Gaussian priors

### S6.1. Model hierarchy and full joint

Let  $\mathbf{X} \in \mathbb{R}^{n \times p}$  have row vectors  $\mathbf{x}_i^\top$ , and let  $\boldsymbol{\beta} \in \mathbb{R}^p$ . The ridge/Gaussian coefficient prior is

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{b}_{\beta,0}, \mathbf{V}_{\beta,0}), \quad \mathbf{b}_{\beta,0} \in \mathbb{R}^p, \quad \mathbf{V}_{\beta,0} \in \mathbb{R}^{p \times p}. \quad (\text{S66})$$

In the package interface, the same Gaussian update is used for `beta_prior = "ridge"`. When `beta_prior = "rhs_ns"`, the diagonal prior precision in this update is replaced by the RHS-NS expected precision described in Section S7. Thus this supplement covers the Gaussian/ridge prior and the conjugate RHS-NS prior, both for MCMC and LDVB. Under the exAL likelihood,

$$v_i \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \quad s_i \sim \mathcal{N}^+(0, 1), \quad (\text{S67})$$

$$y_i \mid \boldsymbol{\beta}, \sigma, \gamma, v_i, s_i \sim \mathcal{N}(\mathbf{x}_i^\top \boldsymbol{\beta} + C_\gamma \sigma |\gamma| s_i + A_\gamma v_i, \sigma B_\gamma v_i). \quad (\text{S68})$$

The static AL model is obtained by setting  $\gamma = 0$ , removing  $s_i$ , and using  $A_{p0}, B_{p0}$ . The exAL full joint is

$$p(\boldsymbol{\beta}, \sigma, \gamma, \mathbf{v}, \mathbf{s}, \mathbf{y}) = p(\boldsymbol{\beta})p(\sigma)\pi_\gamma(\gamma) \prod_{i=1}^n p(y_i \mid \boldsymbol{\beta}, \sigma, \gamma, v_i, s_i)p(v_i \mid \sigma)p(s_i). \quad (\text{S69})$$

### S6.2. MCMC updates

For fixed  $(\sigma, \gamma, \mathbf{v}, \mathbf{s})$ , define

$$y_i^* = y_i - C_\gamma \sigma |\gamma| s_i - A_\gamma v_i, \quad w_i = (\sigma B_\gamma v_i)^{-1}. \quad (\text{S70})$$

Let  $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$  and  $\mathbf{y}^* = (y_1^*, \dots, y_n^*)^\top$ . Then

$$\boldsymbol{\Sigma}_{\beta|\cdot} = \left( \mathbf{V}_{\beta,0}^{-1} + \mathbf{X}^\top \mathbf{W} \mathbf{X} \right)^{-1}, \quad (\text{S71})$$

$$\mathbf{m}_{\beta|\cdot} = \boldsymbol{\Sigma}_{\beta|\cdot} \left( \mathbf{V}_{\beta,0}^{-1} \mathbf{b}_{\beta,0} + \mathbf{X}^\top \mathbf{W} \mathbf{y}^* \right), \quad (\text{S72})$$

$$\boldsymbol{\beta} \mid \cdot \sim \mathcal{N}(\mathbf{m}_{\beta|\cdot}, \boldsymbol{\Sigma}_{\beta|\cdot}). \quad (\text{S73})$$

Let  $D_\gamma = C_\gamma |\gamma|$ . The remaining exAL full conditionals are

$$s_i \mid \cdot \sim \mathcal{N}^+(m_{s,i}, V_{s,i}), \quad (\text{S74})$$

$$V_{s,i} = \left(1 + \frac{\sigma D_\gamma^2}{B_\gamma v_i}\right)^{-1}, \quad (\text{S75})$$

$$m_{s,i} = V_{s,i} \frac{D_\gamma (y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i)}{B_\gamma v_i}, \quad (\text{S76})$$

and, with  $r_i = y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - \sigma D_\gamma s_i$ ,

$$v_i \mid \cdot \sim \text{GIG} \left( \frac{1}{2}, \frac{r_i^2}{\sigma B_\gamma}, \frac{A_\gamma^2}{\sigma B_\gamma} + \frac{2}{\sigma} \right). \quad (\text{S77})$$

For fixed  $\gamma$ , define  $t_i = y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i$ . Then

$$\sigma \mid \cdot \sim \text{GIG} \left( -a_\sigma - \frac{3n}{2}, 2b_\sigma + 2 \sum_{i=1}^n v_i + \sum_{i=1}^n \frac{t_i^2}{B_\gamma v_i}, \sum_{i=1}^n \frac{D_\gamma^2 s_i^2}{B_\gamma v_i} \right). \quad (\text{S78})$$

The nonstandard  $\gamma$  kernel is

$$\pi(\gamma \mid \cdot) \propto \pi_\gamma(\gamma) [B_\gamma]^{-n/2} \exp \left\{ \sum_{i=1}^n \frac{D_\gamma s_i t_i}{B_\gamma v_i} - \frac{1}{2\sigma} \sum_{i=1}^n \frac{t_i^2}{B_\gamma v_i} - \frac{\sigma}{2} \sum_{i=1}^n \frac{D_\gamma^2 s_i^2}{B_\gamma v_i} \right\}, \quad (\text{S79})$$

on  $(L, U)$ . The package default updates this one-dimensional target by the bounded slice sampler in Section S9.2.

For the AL special case, the Gibbs updates reduce to

$$\boldsymbol{\beta} \mid \cdot \sim \mathcal{N}(\mathbf{m}_{\boldsymbol{\beta}|\cdot}, \boldsymbol{\Sigma}_{\boldsymbol{\beta}|\cdot}), \quad (\text{S80})$$

$$\sigma \mid \cdot \sim \text{IG} \left( a_\sigma + \frac{3n}{2}, b_\sigma + \sum_{i=1}^n v_i + \sum_{i=1}^n \frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_{p_0} v_i)^2}{2B_{p_0} v_i} \right), \quad (\text{S81})$$

$$v_i \mid \cdot \sim \text{GIG} \left( \frac{1}{2}, \frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{\sigma B_{p_0}}, \frac{A_{p_0}^2}{\sigma B_{p_0}} + \frac{2}{\sigma} \right). \quad (\text{S82})$$

#### Algorithm S6 (static AL/exAL MCMC).

- (1) Initialize  $\boldsymbol{\beta}$ ,  $\mathbf{v}$ ,  $\mathbf{s}$ ,  $\sigma$ , and  $\gamma$ ; omit  $\mathbf{s}$  and fix  $\gamma = 0$  for the AL path.
- (2) For each MCMC iteration:
  - (a) sample  $\boldsymbol{\beta}$  from (S73);
  - (b) sample  $v_i$  for  $i = 1, \dots, n$ ;
  - (c) for exAL, sample  $s_i$  for  $i = 1, \dots, n$ ;
  - (d) sample  $\sigma$ ;
  - (e) for exAL, sample  $\gamma$  by bounded slice sampling;
  - (f) if `beta_prior = "rhs_ns"`, update the RHS-NS scale block in Section S7;

(g) retain draws after burn-in.

(3) Return coefficient, latent-variable, and scale posterior samples.

### S6.3. Laplace-Delta VB updates and ELBO

The static exAL variational family is

$$q(\boldsymbol{\beta}, \mathbf{v}, \mathbf{s}, \sigma, \gamma) = q(\boldsymbol{\beta}) \prod_{i=1}^n q(v_i) \prod_{i=1}^n q(s_i) q(\sigma, \gamma). \quad (\text{S83})$$

The coefficient factor is Gaussian. With the same moment definitions as in (S48),

$$\omega_i = \kappa_1 \mathbb{E}(v_i^{-1}), \quad (\text{S84})$$

$$\eta_i = y_i \kappa_1 \mathbb{E}(v_i^{-1}) - \kappa_4 \mathbb{E}(v_i^{-1}) \mathbb{E}(s_i) - \kappa_2, \quad (\text{S85})$$

$$\boldsymbol{\Sigma}_\beta = \left( \mathbf{V}_{\beta,0}^{-1} + \mathbf{X}^\top \text{diag}(\omega_1, \dots, \omega_n) \mathbf{X} \right)^{-1}, \quad (\text{S86})$$

$$\mathbf{m}_\beta = \boldsymbol{\Sigma}_\beta \left( \mathbf{V}_{\beta,0}^{-1} \mathbf{b}_{\beta,0} + \mathbf{X}^\top \boldsymbol{\eta} \right), \quad (\text{S87})$$

where  $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)^\top$ . The  $q(v_i)$  and  $q(s_i)$  updates are also closed form conditional on moments of  $q(\sigma, \gamma)$ . Let

$$\bar{r}_i = y_i - \mathbf{x}_i^\top \mathbf{m}_\beta, \quad \bar{r}_i^2 = \bar{r}_i^2 + \mathbf{x}_i^\top \boldsymbol{\Sigma}_\beta \mathbf{x}_i. \quad (\text{S88})$$

Then

$$q(v_i) = \text{GIG} \left( \frac{1}{2}, \kappa_1 \bar{r}_i^2 - 2\kappa_4 \bar{r}_i \mathbb{E}(s_i) + \kappa_5 \mathbb{E}(s_i^2), \kappa_3 + 2\mathbb{E}(\sigma^{-1}) \right), \quad (\text{S89})$$

$$q(s_i) = \mathcal{N}^+(m_{s,i}^{\text{VB}}, V_{s,i}^{\text{VB}}), \quad (\text{S90})$$

$$V_{s,i}^{\text{VB}} = \left( 1 + \kappa_5 \mathbb{E}(v_i^{-1}) \right)^{-1}, \quad (\text{S91})$$

$$m_{s,i}^{\text{VB}} = V_{s,i}^{\text{VB}} \left\{ \kappa_4 \bar{r}_i \mathbb{E}(v_i^{-1}) - \kappa_6 \right\}. \quad (\text{S92})$$

The nonconjugate factor  $q(\sigma, \gamma)$  is proportional to

$$q(\sigma, \gamma) \propto \pi_\gamma(\gamma) [B_\gamma]^{-n/2} \sigma^{-(a_\sigma + 1 + 3n/2)} \exp \left\{ -\frac{1}{2} \left( \frac{\bar{c}(\gamma)}{\sigma} + \bar{d}(\gamma) \sigma \right) + \bar{e}(\gamma) \right\}, \quad (\text{S93})$$

where

$$\bar{c}(\gamma) = 2b_\sigma + 2 \sum_{i=1}^n \mathbb{E}(v_i) + \sum_{i=1}^n \frac{\mathbb{E}\{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i)^2 / v_i\}}{B_\gamma}, \quad (\text{S94})$$

$$\bar{d}(\gamma) = \sum_{i=1}^n \frac{C_\gamma^2 \gamma^2}{B_\gamma} \mathbb{E}\{s_i^2 / v_i\}, \quad (\text{S95})$$

$$\bar{e}(\gamma) = \sum_{i=1}^n \frac{C_\gamma |\gamma|}{B_\gamma} \mathbb{E}\{s_i (y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i) / v_i\}. \quad (\text{S96})$$

This factor is approximated by the same Laplace-Delta method as in the dynamic exDQLM. The AL path sets  $\gamma = 0$  and uses  $q(\boldsymbol{\beta})q(\sigma) \prod_i q(v_i)$  with the same closed-form updates as Section S3, replacing state smoothing moments by  $\mathbb{E}\{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2\}$ .

**Algorithm S7 (static AL/exAL LDVB).**

- (1) Initialize  $q(\beta)$ , the latent-variable factors, the scale factors, and the  $(\sigma, \gamma)$  block.
- (2) Repeat until convergence:
  - (a) update  $q(\beta)$  using (S87);
  - (b) update  $q(v_i)$  and, for exAL,  $q(s_i)$  using (S92);
  - (c) update  $q(\sigma)$  for AL or  $q(\sigma, \gamma)$  for exAL;
  - (d) if `beta_prior = "rhs_ns"`, update the RHS-NS factors in Section S7;
  - (e) monitor moment changes and  $\mathcal{L}(q)$ .
- (3) Draw approximate posterior samples when requested.

The static exAL ELBO is computed from the same blocks as the dynamic exDQLM, with the Gaussian state factor replaced by the Gaussian coefficient factor. Let  $\bar{\ell}_\sigma = \mathbb{E}(\log \sigma)$ ,  $\bar{\ell}_B = \mathbb{E}(\log B_\gamma)$ ,  $\bar{\ell}_\pi = \mathbb{E}\{\log \pi_\gamma(\gamma)\}$ ,  $m_{v,i} = \mathbb{E}(v_i)$ ,  $m_{1/v,i} = \mathbb{E}(v_i^{-1})$ ,  $m_{\log v,i} = \mathbb{E}(\log v_i)$ ,  $m_{s,i} = \mathbb{E}(s_i)$ , and  $m_{s^2,i} = \mathbb{E}(s_i^2)$ . Also define

$$t_i = y_i - \mathbf{x}_i^\top \mathbf{m}_\beta, \quad q_i = \mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_i. \quad (\text{S97})$$

Then

$$\begin{aligned} \mathcal{L}(q) = & \mathcal{L}_{\text{like}} + \mathcal{L}_v + \mathcal{L}_s + \mathcal{L}_\beta + \mathcal{L}_{\sigma\gamma} + H\{q(\beta)\} + \sum_{i=1}^n H\{q(v_i)\} \\ & + \sum_{i=1}^n H\{q(s_i)\} + H\{q(\sigma, \gamma)\}, \end{aligned} \quad (\text{S98})$$

where

$$\begin{aligned} \mathcal{L}_{\text{like}} = & -\frac{n}{2} \log(2\pi) - \frac{n}{2} (\bar{\ell}_B + \bar{\ell}_\sigma) - \frac{1}{2} \sum_{i=1}^n m_{\log v,i} \\ & - \frac{1}{2} \sum_{i=1}^n \left\{ \kappa_1 m_{1/v,i} (t_i^2 + q_i) - 2\kappa_2 t_i + \kappa_3 m_{v,i} \right\} \\ & + \sum_{i=1}^n \left\{ \kappa_4 m_{s,i} m_{1/v,i} t_i - \kappa_6 m_{s,i} - \frac{1}{2} \kappa_5 m_{s^2,i} m_{1/v,i} \right\}, \end{aligned} \quad (\text{S99})$$

$$\mathcal{L}_v = -n\bar{\ell}_\sigma - \mathbb{E}(\sigma^{-1}) \sum_{i=1}^n m_{v,i}, \quad (\text{S100})$$

$$\mathcal{L}_s = n \log 2 - \frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n m_{s^2,i}, \quad (\text{S101})$$

$$\mathcal{L}_{\sigma\gamma} = a_\sigma \log b_\sigma - \log \Gamma(a_\sigma) - (a_\sigma + 1) \bar{\ell}_\sigma - b_\sigma \mathbb{E}(\sigma^{-1}) + \bar{\ell}_\pi, \quad (\text{S102})$$

$$\mathcal{L}_\beta = \mathbb{E}_q\{\log p(\beta)\}. \quad (\text{S103})$$

For the ridge prior,  $\mathcal{L}_\beta$  is the Gaussian prior expectation. For the RHS-NS prior, it is replaced by the RHS-NS contribution in Section S7. The Gaussian, GIG, truncated-normal, and

Laplace-Delta entropy terms are given by (S144), (S147), (S148), and (S150), respectively. The AL path is recovered by setting  $\gamma = 0$  and dropping the  $s_i$  and  $(\sigma, \gamma)$  Laplace-Delta blocks; then  $q(\sigma)$  is inverse-gamma and uses (S146).

## S7. Static regression with the RHS-NS prior

This section focuses on the `rhs_ns` coefficient prior, the Gibbs-friendly regularized horseshoe variant used for sparse static regression. It enters the static Gaussian update for  $\beta$  through a prior precision matrix. Let  $\mathcal{A} \subseteq \{1, \dots, p\}$  denote the active shrinkage set; by default an intercept, if present, is not shrunk. For  $j \notin \mathcal{A}$  the package uses a fixed intercept precision. The direct `rhs` path is not developed here because its log-scale block is nonconjugate; the `rhs_ns` path is the conjugate sparse-prior formulation documented in this supplement.

### S7.1. RHS-NS prior hierarchy

For  $j \in \mathcal{A}$ , the `rhs_ns` hierarchy is

$$\beta_j \mid \tau^2, \lambda_j^2, \zeta^2 \propto \mathcal{N}(\beta_j \mid 0, \tau^2 \lambda_j^2) \mathcal{N}(\beta_j \mid 0, \zeta^2), \quad (\text{S104})$$

$$\lambda_j^2 \mid \nu_j \sim \text{IG}(1/2, 1/\nu_j), \quad \nu_j \sim \text{IG}(1/2, 1), \quad (\text{S105})$$

$$\tau^2 \mid \xi \sim \text{IG}(1/2, 1/\xi), \quad \xi \sim \text{IG}(1/2, 1/\tau_0^2), \quad (\text{S106})$$

$$\zeta^2 \sim \text{IG}(a_\zeta, b_\zeta), \quad (\text{S107})$$

unless  $\zeta^2$  is fixed by the user. The implementation includes the scale-dependent Gaussian log-normalizing terms shown in the ELBO contribution below. Thus the proportionality in (S107) denotes the coefficient kernel, while the surrounding scale updates and ELBO use the corresponding normal-density contributions. The resulting prior precision contribution for active coefficients is

$$\omega_j^{\text{prior}} = \frac{1}{\tau^2 \lambda_j^2} + \frac{1}{\zeta^2}. \quad (\text{S108})$$

Consequently, the  $\beta$  update in MCMC or VB is obtained from the static Gaussian update after replacing  $\mathbf{V}_{\beta,0}^{-1}$  by a diagonal precision matrix with active entries  $\omega_j^{\text{prior}}$  or their variational expectations.

### S7.2. RHS-NS MCMC updates

Conditioning on  $\beta$ , the shrinkage block has closed-form inverse-gamma updates:

$$\lambda_j^2 \mid \cdot \sim \text{IG} \left( 1, \frac{1}{\nu_j} + \frac{\beta_j^2}{2\tau^2} \right), \quad (\text{S109})$$

$$\nu_j \mid \cdot \sim \text{IG} \left( 1, 1 + \frac{1}{\lambda_j^2} \right), \quad (\text{S110})$$

$$\tau^2 \mid \cdot \sim \text{IG} \left( \frac{|\mathcal{A}|+1}{2}, \frac{1}{\xi} + \frac{1}{2} \sum_{j \in \mathcal{A}} \frac{\beta_j^2}{\lambda_j^2} \right), \quad (\text{S111})$$

$$\xi \mid \cdot \sim \text{IG} \left( 1, \frac{1}{\tau_0^2} + \frac{1}{\tau^2} \right), \quad (\text{S112})$$

$$\zeta^2 \mid \cdot \sim \text{IG} \left( a_\zeta + \frac{|\mathcal{A}|}{2}, b_\zeta + \frac{1}{2} \sum_{j \in \mathcal{A}} \beta_j^2 \right), \quad (\text{S113})$$

with the last line omitted when  $\zeta^2$  is fixed. These equations are the conditional targets for the RHS-NS scale block. The implementation may temporarily freeze or schedule the global-scale update during configured warmup iterations for numerical stability; this changes only the update schedule, not the conditional target displayed above.

### S7.3. RHS-NS VB updates and ELBO contribution

The package uses a mean-field approximation over the RHS-NS scale block. For  $j \in \mathcal{A}$ ,

$$q(\lambda_j^2) = \text{IG}(a_{\lambda_j}, b_{\lambda_j}), \quad q(\nu_j) = \text{IG}(a_{\nu_j}, b_{\nu_j}), \quad (\text{S114})$$

$$q(\tau^2) = \text{IG}(a_\tau, b_\tau), \quad q(\xi) = \text{IG}(a_\xi, b_\xi), \quad (\text{S115})$$

$$q(\zeta^2) = \text{IG}(a_\zeta^*, b_\zeta^*), \quad (\text{S116})$$

with  $a_{\lambda_j} = a_{\nu_j} = a_\xi = 1$  and  $a_\tau = (|\mathcal{A}| + 1)/2$ . Let  $\bar{\beta}_j^2 = \mathbb{E}_q(\beta_j^2)$ . The coordinate updates are

$$b_{\lambda_j} = \frac{1}{2} \bar{\beta}_j^2 \mathbb{E}(\tau^{-2}) + \mathbb{E}(\nu_j^{-1}), \quad (\text{S117})$$

$$b_{\nu_j} = 1 + \mathbb{E}\{(\lambda_j^2)^{-1}\}, \quad (\text{S118})$$

$$b_\tau = \frac{1}{2} \sum_{j \in \mathcal{A}} \bar{\beta}_j^2 \mathbb{E}\{(\lambda_j^2)^{-1}\} + \mathbb{E}(\xi^{-1}), \quad (\text{S119})$$

$$b_\xi = \frac{1}{\tau_0^2} + \mathbb{E}(\tau^{-2}), \quad (\text{S120})$$

$$a_\zeta^* = a_\zeta + \frac{|\mathcal{A}|}{2}, \quad b_\zeta^* = b_\zeta + \frac{1}{2} \sum_{j \in \mathcal{A}} \bar{\beta}_j^2. \quad (\text{S121})$$

The third line is shorthand for  $b_\tau = \frac{1}{2} \sum_{j \in \mathcal{A}} \bar{\beta}_j^2 \mathbb{E}\{(\lambda_j^2)^{-1}\} + \mathbb{E}(\xi^{-1})$ , where the expectation of  $(\lambda_j^2)^{-1}$  is component-specific inside the summation. As in MCMC, optional warmup scheduling for the  $\tau^2, \xi$  block affects when the coordinate is updated, not the coordinate target itself.



The following term-by-term ELBO contribution matches the package's `rhs_ns` scale block. For  $x \sim \text{IG}(a, b)$  under the rate parameterization used above, define

$$m_{\log x}(a, b) = \log b - \psi(a), \quad m_{x^{-1}}(a, b) = a/b, \quad (\text{S122})$$

$$h_{\text{IG}}(a, b) = a + \log b + \log \Gamma(a) - (a + 1)\psi(a), \quad (\text{S123})$$

where  $\psi(\cdot)$  is the digamma function and  $h_{\text{IG}}$  is the entropy of an inverse-gamma variational factor. Let

$$\ell_{\lambda j} = m_{\log x}(a_{\lambda j}, b_{\lambda j}), \quad r_{\lambda j} = m_{x^{-1}}(a_{\lambda j}, b_{\lambda j}), \quad (\text{S124})$$

$$\ell_{\nu j} = m_{\log x}(a_{\nu j}, b_{\nu j}), \quad r_{\nu j} = m_{x^{-1}}(a_{\nu j}, b_{\nu j}), \quad (\text{S125})$$

$$\ell_{\tau} = m_{\log x}(a_{\tau}, b_{\tau}), \quad r_{\tau} = m_{x^{-1}}(a_{\tau}, b_{\tau}), \quad (\text{S126})$$

$$\ell_{\xi} = m_{\log x}(a_{\xi}, b_{\xi}), \quad r_{\xi} = m_{x^{-1}}(a_{\xi}, b_{\xi}). \quad (\text{S127})$$

If  $\zeta^2$  is random, set  $\ell_{\zeta} = m_{\log x}(a_{\zeta}^*, b_{\zeta}^*)$  and  $r_{\zeta} = m_{x^{-1}}(a_{\zeta}^*, b_{\zeta}^*)$ ; if it is fixed at  $\zeta_0^2$ , set  $\ell_{\zeta} = \log \zeta_0^2$  and  $r_{\zeta} = 1/\zeta_0^2$ .

The coefficient-prior contribution is

$$\mathcal{L}_{\beta, \text{hs}} = \sum_{j \in \mathcal{A}} \left\{ -\frac{1}{2} \log(2\pi) - \frac{1}{2} \ell_{\tau} - \frac{1}{2} \ell_{\lambda j} - \frac{1}{2} \bar{\beta}_j^2 r_{\tau} r_{\lambda j} \right\}, \quad (\text{S128})$$

$$\mathcal{L}_{\beta, \text{slab}} = \sum_{j \in \mathcal{A}} \left\{ -\frac{1}{2} \log(2\pi) - \frac{1}{2} \ell_{\zeta} - \frac{1}{2} \bar{\beta}_j^2 r_{\zeta} \right\}. \quad (\text{S129})$$

The local, global, and slab prior contributions are

$$\mathcal{L}_{\lambda|\nu} = \sum_{j \in \mathcal{A}} \left\{ -\frac{1}{2} \ell_{\nu j} - \log \Gamma(1/2) - \frac{3}{2} \ell_{\lambda j} - r_{\nu j} r_{\lambda j} \right\}, \quad (\text{S130})$$

$$\mathcal{L}_{\nu} = \sum_{j \in \mathcal{A}} \left\{ -\log \Gamma(1/2) - \frac{3}{2} \ell_{\nu j} - r_{\nu j} \right\}, \quad (\text{S131})$$

$$\mathcal{L}_{\tau|\xi} = -\frac{1}{2} \ell_{\xi} - \log \Gamma(1/2) - \frac{3}{2} \ell_{\tau} - r_{\xi} r_{\tau}, \quad (\text{S132})$$

$$\mathcal{L}_{\xi} = \frac{1}{2} \log(\tau_0^{-2}) - \log \Gamma(1/2) - \frac{3}{2} \ell_{\xi} - \tau_0^{-2} r_{\xi}, \quad (\text{S133})$$

$$\mathcal{L}_{\zeta} = \begin{cases} a_{\zeta} \log b_{\zeta} - \log \Gamma(a_{\zeta}) - (a_{\zeta} + 1)\ell_{\zeta} - b_{\zeta} r_{\zeta}, & \zeta^2 \text{ random,} \\ 0, & \zeta^2 \text{ fixed.} \end{cases} \quad (\text{S134})$$

The entropy contribution is

$$\mathcal{L}_{\text{ent}} = \sum_{j \in \mathcal{A}} h_{\text{IG}}(a_{\lambda j}, b_{\lambda j}) + \sum_{j \in \mathcal{A}} h_{\text{IG}}(a_{\nu j}, b_{\nu j}) + h_{\text{IG}}(a_{\tau}, b_{\tau}) + h_{\text{IG}}(a_{\xi}, b_{\xi}) + \mathbb{I}(\zeta^2 \text{ random}) h_{\text{IG}}(a_{\zeta}^*, b_{\zeta}^*). \quad (\text{S135})$$

Finally, when the intercept is not included in  $\mathcal{A}$ , the package adds the fixed Gaussian intercept-prior contribution

$$\mathcal{L}_{\beta_0} = \frac{1}{2} \{\log(\pi_0) - \log(2\pi)\} - \frac{1}{2} \pi_0 \bar{\beta}_0^2, \quad (\text{S136})$$

where  $\pi_0$  is the fixed intercept precision. If the intercept is shrunk, this term is omitted because the intercept is already represented in the active RHS-NS block.

Thus the package contribution is

$$\mathcal{L}_{\text{rhs\_ns}} = \mathcal{L}_{\beta, \text{hs}} + \mathcal{L}_{\beta, \text{slab}} + \mathcal{L}_{\lambda|\nu} + \mathcal{L}_{\nu} + \mathcal{L}_{\tau|\xi} + \mathcal{L}_{\xi} + \mathcal{L}_{\zeta} + \mathcal{L}_{\text{ent}} + \mathcal{L}_{\beta_0}. \quad (\text{S137})$$

All terms are available in closed form because each scale factor is inverse-gamma. This is the RHS-NS component of the full static-model ELBO; the likelihood, latent-variable, and  $(\sigma, \gamma)$  terms are added by the surrounding static AL or exAL variational routine.

## S8. Forecasting, diagnostics, and posterior-predictive synthesis

### S8.1. Forecasting

For a posterior draw  $m$  at forecast origin  $\tilde{t}$ , the state forecast recursion in the main article gives

$$\theta_{\tilde{t}+k}^{(m)} \sim \mathcal{N}\{a_{\tilde{t}}^{(m)}(k), R_{\tilde{t}}^{(m)}(k)\}, \quad k = 1, \dots, K. \quad (\text{S138})$$

Given a forecasted state draw, posterior predictive draws are generated from

$$Y_{\tilde{t}+k}^{\text{rep},(m)} \sim \text{exAL}_{p_0}\{\mathbf{F}_{\tilde{t}+k}^{\top} \theta_{\tilde{t}+k}^{(m)}, \sigma^{(m)}, \gamma^{(m)}\}. \quad (\text{S139})$$

For the AL/DQLM special case, set  $\gamma^{(m)} = 0$ . The function `exdqlmForecast()` implements this recursion for MCMC and VB fitted objects; for VB fits, the posterior draws are replaced by draws from the fitted variational factors.

### S8.2. Diagnostics

The dynamic diagnostics use the MAP one-step-ahead standardized forecast-error sequence produced by the filtering recursion. If  $\hat{f}_t$  and  $\hat{q}_t$  denote the fitted one-step-ahead location and variance used in this plug-in sequence, the diagnostic error and its normal-score transform are

$$\hat{e}_t = \frac{y_t - \hat{f}_t}{\sqrt{\hat{q}_t}}, \quad \hat{u}_t = \Phi(\hat{e}_t). \quad (\text{S140})$$

The QQ plot, ACF plot, KL divergence, and standardized forecast-error plot are computed from  $\{\hat{e}_t\}$  or  $\{\hat{u}_t\}$  rather than from posterior uncertainty bands for residuals.

For predictive draws  $y_t^{(1)}, \dots, y_t^{(M)}$ , the empirical CRPS estimator is

$$\widehat{\text{CRPS}}_t = \frac{1}{M} \sum_{m=1}^M |y_t^{(m)} - y_t^{\text{obs}}| - \frac{1}{2M^2} \sum_{m=1}^M \sum_{\ell=1}^M |y_t^{(m)} - y_t^{(\ell)}|. \quad (\text{S141})$$

The reported CRPS is the average of (S141) over the diagnostic time points (Gneiting and Raftery 2007). For target quantile  $p_0$ , the posterior predictive loss criterion used by the package is

$$\widehat{\text{pplc}} = \sum_t \frac{1}{M} \sum_{m=1}^M \rho_{p_0}\{y_t^{\text{obs}} - y_t^{\text{rep},(m)}\}, \quad (\text{S142})$$

where  $\rho_{p_0}$  is the check-loss function (Gelfand and Ghosh 1998). The dynamic diagnostic routine returns these summaries. The static diagnostic routine returns check-loss summaries and reference-quantile errors when a reference quantile is supplied.

### S8.3. Posterior-predictive synthesis

The function `quantileSynthesis()` combines posterior predictive draws from separately fitted models at ordered levels  $p_1 < \dots < p_K$ . For each time point and draw index, it forms the quantile-specific predictive curve, applies the requested monotonicity correction when fitted quantiles cross, and interpolates across  $p$  to obtain draws from one posterior predictive distribution. The monotonicity correction uses isotonic adjustment and, when requested, global monotone rearrangement (Barlow, Bartholomew, Bremner, and Brunk 1972; Chernozhukov, Fernández-Val, and Galichon 2010).

#### Algorithm S8 (posterior-predictive synthesis).

- (1) Collect posterior predictive draw matrices from fitted models at  $p_1 < \dots < p_K$ .
- (2) For each time point and draw index, form the quantile-specific predictive curve.
- (3) Apply the requested monotonicity correction.
- (4) Interpolate across quantile levels.
- (5) Return synthesized posterior predictive draws and summaries.

## S9. Numerical and ELBO implementation blocks

### S9.1. Reusable moment and entropy formulas

The VB routines monitor

$$\mathcal{L}(q) = \mathbb{E}_q\{\log p(\text{complete data})\} + H(q), \quad H(q) = -\mathbb{E}_q\{\log q\}. \quad (\text{S143})$$

The following moment and entropy formulas are reused by the dynamic and static algorithms.

If  $\mathbf{x} \sim \mathcal{N}(\mathbf{m}, \mathbf{V})$  with  $\mathbf{x} \in \mathbb{R}^d$ , then

$$H\{\mathcal{N}(\mathbf{m}, \mathbf{V})\} = \frac{1}{2}\{d(1 + \log 2\pi) + \log |\mathbf{V}|\}. \quad (\text{S144})$$

If  $x \sim \text{IG}(a, b)$  under the parameterization in Section S2, then

$$\mathbb{E}(x^{-1}) = \frac{a}{b}, \quad \mathbb{E}(\log x) = \log b - \psi(a), \quad (\text{S145})$$

and

$$H\{\text{IG}(a, b)\} = a + \log b + \log \Gamma(a) - (a + 1)\psi(a), \quad (\text{S146})$$

where  $\psi(\cdot)$  is the digamma function.

If  $x \sim \text{GIG}(\lambda, \chi, \psi_{\text{GIG}})$  and  $z = (\chi\psi_{\text{GIG}})^{1/2}$ , let  $m_x = \mathbb{E}(x)$ ,  $m_{1/x} = \mathbb{E}(x^{-1})$ , and  $m_{\log x} = \mathbb{E}(\log x)$ . Then

$$\begin{aligned} H\{\text{GIG}(\lambda, \chi, \psi_{\text{GIG}})\} &= \log\{2K_\lambda(z)\} - \frac{\lambda}{2} \log\left(\frac{\psi_{\text{GIG}}}{\chi}\right) - (\lambda - 1)m_{\log x} \\ &\quad + \frac{1}{2}\{\chi m_{1/x} + \psi_{\text{GIG}} m_x\}. \end{aligned} \quad (\text{S147})$$

The moments  $m_x$  and  $m_{1/x}$  follow from (S11);  $m_{\log x}$  is the derivative of  $\log K_\lambda(z)$  with respect to  $\lambda$  plus  $\frac{1}{2} \log(\chi/\psi_{\text{GIG}})$ , and is evaluated numerically in the package.

If  $x \sim \mathcal{N}^+(m, V)$ , let  $a = V^{1/2}$ ,  $z = m/a$ , and  $\Lambda(z) = \phi(z)/\Phi(z)$ . The entropy is

$$H\{\mathcal{N}^+(m, V)\} = \frac{1}{2} \log(2\pi V) + \log \Phi(z) + \frac{1}{2} \{1 - z\Lambda(z)\}. \quad (\text{S148})$$

This closed form is used for the truncated-normal latent factors; numerically, the package evaluates the required moments using stable tail-probability guards.

For exAL VB, the nonconjugate  $(\sigma, \gamma)$  block is handled on transformed coordinates

$$\ell_\sigma = \log \sigma, \quad z_\gamma = \text{logit} \left( \frac{\gamma - L}{U - L} \right). \quad (\text{S149})$$

If  $\mathbf{z} = (z_\gamma, \ell_\sigma)^\top$  has Laplace approximation  $q_z(\mathbf{z}) = \mathcal{N}(\hat{\mathbf{z}}, \Sigma_z)$  and  $(\gamma, \sigma) = T(\mathbf{z})$ , then the entropy contribution on the original scale is approximated as

$$H\{q(\sigma, \gamma)\} \approx \frac{1}{2} \{2(1 + \log 2\pi) + \log |\Sigma_z| + \mathbb{E}_{q_z} \{\log |J_T(\mathbf{z})|\}\}, \quad (\text{S150})$$

where

$$\log |J_T(\mathbf{z})| = \ell_\sigma + \log(U - L) + \log r + \log(1 - r), \quad r = \text{expit}(z_\gamma). \quad (\text{S151})$$

The final expectation in (S150), and other smooth expectations of functions of  $(\sigma, \gamma)$ , are evaluated by the same second-order Delta approximation used in the LDVB updates.

## S9.2. Bounded univariate slice sampler

Let  $h(x)$  be an unnormalized log-density on a finite interval  $[a, b]$ , and let  $x_0 \in (a, b)$ . The bounded slice update implemented in the package is:

- (1) Draw  $e \sim \text{Exp}(1)$  and set the log-slice height  $z = h(x_0) - e$ .
- (2) Draw  $u \sim \text{Unif}(0, w)$  and initialize  $L = x_0 - u$ ,  $R = x_0 + w - u$ .
- (3) Step out left and right by increments of  $w$ , respecting the bounds  $a$  and  $b$ , until the interval endpoints lie below the slice or the maximum number of expansions is reached.
- (4) Truncate the interval to  $[a, b]$ .
- (5) Repeatedly draw  $x_1 \sim \text{Unif}(L, R)$ . If  $h(x_1) \geq z$ , accept  $x_1$ . Otherwise shrink the interval to  $[L, x_1]$  if  $x_1 > x_0$  and to  $[x_1, R]$  otherwise.

For dynamic and static exAL MCMC fits, this sampler is applied to the one-dimensional  $\gamma$  target on  $(L, U)$  by default.

## S9.3. Laplace-Delta approximation

For a nonconjugate block  $\boldsymbol{\eta} \in \mathbb{R}^d$  with log target  $\ell(\boldsymbol{\eta})$ , let  $\hat{\boldsymbol{\eta}}$  be a local mode and  $\mathbf{H} = \nabla^2 \ell(\hat{\boldsymbol{\eta}})$ . If  $-\mathbf{H}$  is positive definite, the Laplace approximation is

$$q(\boldsymbol{\eta}) \approx \mathcal{N}(\hat{\boldsymbol{\eta}}, [-\mathbf{H}]^{-1}). \quad (\text{S152})$$

For a smooth function  $r(\boldsymbol{\eta})$ , the Delta approximation used by LDVB is

$$\mathbb{E}\{r(\boldsymbol{\eta})\} \approx r(\hat{\boldsymbol{\eta}}) + \frac{1}{2} \text{tr} \left[ \nabla^2 r(\hat{\boldsymbol{\eta}}) [-\mathbf{H}]^{-1} \right]. \quad (\text{S153})$$

This is used for smooth functions of  $(\sigma, \gamma)$  in dynamic and static exAL models.

## S10. Package-to-algorithm map

Sections S3–S9 define the targets used by the exported fitting and post-processing functions. Some routines temporarily hold or schedule selected blocks during early iterations for numerical stability. These controls affect update timing only; they do not change the posterior target, full conditional distributions, or coordinate optima listed in this supplement.

| Function                                    | Section | Derivation or computational block                       |
|---------------------------------------------|---------|---------------------------------------------------------|
| <code>exdqlmMCMC()</code>                   | S3–S4   | Dynamic DQLM/exDQLM MCMC; Algorithms S1 and S3.         |
| <code>exdqlmLDVB()</code>                   | S3–S4   | Dynamic DQLM/exDQLM LDVB; Algorithms S2 and S4.         |
| <code>exdqlmTransferMCMC()</code>           | S5      | Transfer-function MCMC; Algorithm S5 with Algorithm S3. |
| <code>exdqlmTransferLDVB()</code>           | S5      | Transfer-function LDVB; Algorithm S5 with Algorithm S4. |
| <code>exalStaticMCMC()</code>               | S6–S7   | Static AL/exAL MCMC; Algorithm S6.                      |
| <code>exalStaticLDVB()</code>               | S6–S7   | Static AL/exAL LDVB; Algorithm S7.                      |
| <code>exdqlmForecast()</code>               | S8.1    | Forecast recursion and predictive draws.                |
| <code>exdqlmDiagnostics()</code>            | S8.2    | Dynamic PIT, CRPS, and ppls diagnostics.                |
| <code>exalStaticDiagnostics()</code>        | S8.2    | Static check-loss and reference-quantile diagnostics.   |
| <code>quantileSynthesis()</code>            | S8.3    | Posterior-predictive synthesis; Algorithm S8.           |
| <code>dexal()</code> , <code>pexal()</code> | S2      | exAL distribution evaluation and simulation.            |
| <code>qexal()</code> , <code>rexal()</code> |         |                                                         |
| <code>get_gamma_bounds()</code>             | S2      | Roots $L, U$ for a fixed $p_0$ .                        |

Table S3: Mapping between exported **exdqlm** functions and the algorithmic blocks derived in this supplement.

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